

Hypotheses, underdetermination, and wrong models

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Abstract We reanalyse the corpus data of Walkden et al. (2023), which uses three historical corpus studies to assess Trudgill’s theory of sociolinguistic typology, according to which short-term adult-driven language contact leads to simplification effects in language. We argue, on methodological and epistemological grounds, that Studies 1 and 2 provide no evidence for the hypotheses they are taken to support. The problem is twofold. At the level of inference, the paper reads low p-values from Null-Hypothesis Significance Testing as corroboration of a large, loosely specified network of hypotheses, when such a test has almost no capacity to expose those hypotheses as false if they were indeed false. At the level of the data, both studies mishandle aggregation: Study 1 collapses thousands of binary outcomes into a few dozen equally weighted per-text difference scores, and Study 2 uses an analogous aggregate as a regressor. Refitting without these errors, we find that little survives. In Study 1, once texts enter as a random effect, between-text variation swamps dialect, and the one generalisation that holds rests essentially on a single text. In Study 2, the reported model is all but indistinguishable from a model with only a random effect for individual texts and no other explanatory variables at all. We close with three suggested remedies: (i) disciplined, severity-aware inference paired with more numerically committed theories, (ii) open data as a routine part of peer review, and (iii) support for publishing negative findings. More broadly, historical-corpus data of this kind might rarely support a confirmatory framework, and the deeper error is forcing exploratory-grade data through a confirmatory machinery they were never meant to pass through.

1 Families of hypotheses and underdetermination in science

1.1 Testing theories

Finding things out involves testing hypotheses, but the hypothesis a study can actually measure is rarely the one it cares about. We follow the broad rationalist view that theories are models of some reality that must survive serious testing to count as adequate, and that we learn most from the ways they turn out to be wrong (Mayo 1996; Mayo 2009). The difficulty is that a test never confronts a single hypothesis. It confronts a network: the substantive

claim of interest, the particular prediction derived from it, and a cushion of auxiliary assumptions that must hold for the derivation and the measurement to go through. When a prediction fails, any node or any link in that network can be blamed. When it succeeds, the success is shared out among all of them. The remainder of this section reconstructs that network for Walkden et al. (2023) and then asks what kind of support a corpus test of a particular hypothesis can really transmit to the substantive theory above it.

The authors are explicit that the substantive target is Trudgill's theory as advertised in Trudgill (2011) and elsewhere (emphasis ours):

This article is about how to **assess Trudgill's theory** using the most powerful methodological tool that contemporary historical linguistics has at its disposal: the study of the historical record itself, in the form of textual corpora. (Walkden et al. 2023: 546)

1.2 Substantive hypotheses and their auxiliaries

The theory to be evaluated collects **substantive hypotheses**, its tenets, such as (among many others):

- Adults learn second languages through a different mechanism than children.
- The adult mechanism leads to simplification, ideally because of spelled-out cognitive principles.
- The social parameters of the contact situation influence the direction and pace of change.
- This yields different temporal scales at which contact-induced change takes place.

Refuting hypotheses this broad is hard. They rest on the aforementioned cushion of **auxiliary hypotheses** that can always absorb an apparent refutation, and they are almost never tested directly, and they have to be broken down into particular hypotheses, which are in turn converted into statistical ones. The boundary between tenet and auxiliary is itself not always sharp. Some auxiliaries any linguist would grant. Others are substantive in their own right. Again, we merely provide some examples of a much larger set of hypotheses that have to be assumed to be true:

- L2 learners may exert linguistic influence on L1 speakers of the same language.
- Language contact can induce change under certain (ideally fully specified) conditions.
- The theory applies universally because it rests on general cognitive and/or social principles.
- “*Population-level patterns are not entirely random and [...] individuals' usage can be non-categorical but still pattern meaningfully in probabilistic ways*” (Walkden et al. 2023: 549) — a strong and very specific assumption.
- The loss of number inflection on pre-nominal elements is simplification.
- The addition of pronominal subjects in a predominantly pro-drop language is simplification.
- The move from synthetic case inflection to adpositional marking is simplification.

1.3 Particular hypotheses and their auxiliaries

The authors then introduce the historical corpus-based approach as suited to collecting evidence for the theory, which adds a further tier of assumptions about the method's applicability. Because the theory makes **predictions**, they derive **particular hypotheses**. These cannot themselves be tenets, or they would not be predictions. The approach is avowedly confirmatory — the studies are offered as *support*:

In **support of our approach**, we present three case studies of morphosyntactic phenomena for which the theory of sociolinguistic typology makes predictions, and assess them using historical corpus evidence. [The three case studies] taken together [illustrate] how historical corpus evidence can be brought to bear on questions of sociolinguistic typology. (Walkden et al. 2023: 547)

From this and the earlier quote we can deduce that the authors are in fact explicitly trying to achieve two things at once: test Trudgill's theory and verify that their method provides adequate means for doing so. It is not made explicit which of the two neighbourhoods of the network of hypotheses (the methodological or the deep-theory neighbourhood) would be identified as the source of the error in case the test failed. Some concrete particular hypotheses concern the diachronic development of:

- the loss of number marking on pre-nominal elements in English under Norse contact (Section 3.1);
- the rise of overtly expressed pronominal subjects in Latin American Spanish under contact with enslaved Africans who acquired Spanish as adults (Section 3.2);
- the reorganisation of case and adposition systems in the diffuse contact situation of the Balkan sprachbund.

Each carries its own **auxiliary hypotheses** of measurement and operationalisation — not derived from the theory but required for the procedure to mean anything, for example:

- The mechanisms posited by the theory are mirrored reliably, and with negligible delay, in the written record.
- The surviving sources are rich and faithful enough to register subtle diachronic change.
- The change observed is due largely to contact, and not to general tendencies of change that happen to fall in the same place and time.
- The authors of the texts were connected enough to the speech community to be good exponents of the change, rather than silent monks or ivory-tower writers.

1.4 Underdetermination, and why it matters in linguistics

A successful test using corpus data is supposed to transmit support upward, from the particular hypothesis to the substantive theory, by treating every auxiliary as true. But the auxiliaries are not innocent. Any false one can manufacture a pseudo-confirmation. The method may be inadequate to the surviving sources, the loss of pre-nominal inflection may not be “simplification” in any relevant sense, Trudgill's theory may simply be wrong. The advertised reading holds only if the auxiliaries are *true to a satisfying degree* — not merely

plausible, since plausibility measures our belief, and the validity of the inference does not turn on what we believe.

This is the classical problem of the **underdetermination** of theory by data, and it comes in two forms that compound here. Its *holistic* form is the familiar Duhem–Quine point that a refutation falls on the whole conjunction of hypotheses, not the one under test — the form already built into the network of Section 1 (Duhem 1914; Quine 1951). Laudan (1990) argued that this was overstated as in mature sciences we can usually localise the blame and repair the network. That reply is fair where theories make precise predictions, which is exactly where linguistics (including the present case and much of our own work) is substantially different from the examples Laudan probably had in mind.

Contrastive underdetermination compounds the problem. A set of theories is contrastively underdetermined when the data are compatible with all of them, and nothing in the data breaks the tie. Laudan & Leplin (1991) dismissed this on the grounds that two genuine rivals will seldom draw exactly equal support from the same evidence. Their argument, though, presupposes that the rivals make claims precise enough for “more” and “less” support to be measured. However, linguistic theories more often than not do not make such precise claims. Many of them are largely informal and pre-numerical (which does not mean they could never make such precise predictions), so the evidence cannot be apportioned between rivals in the first place. Consider Study 1: the loss of plural marking on every NP element but the head noun is compatible with Norse contact and with the well-known general drift of English away from inflection from Old to Early Modern English, which might have accidentally progressed faster in the Northern regions than in the Southern ones. A North-to-South progression during the Danelaw would tip the balance toward the contact account — but only somewhat, and the data remain compatible with both. There is no numerical prediction sharp enough in current theories to break the tie decisively.

This is what we mean by the loose fit of informal theory to data, and it is the common root of both forms. Holistic underdetermination keeps a failed prediction from refuting the theory, because the blame can always fall on an auxiliary. Contrastive underdetermination keeps a successful one from confirming it, because a rival is compatible with the same result. The imprecision of the theory is what allows each to operate. A circle can be drawn around any figure, if it is large enough. The practical consequence is that, for substantive linguistic hypotheses, we can secure neither confirmation nor refutation. The most a corpus study like the ones we routinely run can deliver is that the data either fail to contradict the conjunction of our hypotheses (which is not confirmation) or fail to be compatible with it (which, given the loose fit, is not clean refutation either).

The asymmetry with the mature sciences is stark. We invite the reader to name one major linguistic theory of the last thirty-odd years that was abandoned because it stood in clear, repeated, demonstrated conflict with data. Theories have gone out of fashion, but that is not the same thing. Compare Newton’s gravitation, ultimately broken by the anomalous precession of Mercury’s perihelion (Le Verrier 1859): the numbers from the theory and the numbers from observation did not match, and there was no way around it. Linguistics, as of today, rarely affords numbers that unforgiving. Keeping this in view is essential for the next section, where we turn to what statistics can and cannot add to the testing of such

hypotheses.

1.5 Statistical hypotheses and severe testing

For quantitative work, the particular hypotheses of the previous section must be turned into **statistical hypotheses**: expectations about a numerical outcome of counting or measuring. Given the low degree of formalisation, that outcome is usually as imprecise as “ n should be greater than m .” For the loss of number marking on pre-nominal elements under Norse contact, for instance, we might expect inflected pre-nominal elements to occur less often in Northern and Eastern dialects than in Southern and Western ones during the Danelaw. Call this H_1 : “*Occurrence counts of inflected pre-nominal elements are lower in the North and East than in the South and West.*”

This is where matters more often than not get confusing. It is statistically far easier to check a sample for (unusually substantial) *divergence* from a hypothesis than for *accordance* with it, so the hypothesis is inverted into a *Null*, H_0 , and frequentist theory gives the probability of obtaining a sample at least as extreme as the observed one if the Null were true – the p-value. A low p-value (by convention usually $p < 0.05$) is then read as licence to infer H_1 , which, if we read the previous sentence carefully, it is not. That approach to statistical inference is called Null-Hypothesis Significance Testing (NHST), and it has been under heavy fire for two decades, almost unnoticed in linguistics.¹ Several aspects of NHST inference are outright invalid (Gigerenzer 2004; Greenland et al. 2016; Mayo 2018), and we only name three. First and most devastatingly, the incompatibility of a single set of data with a rather uninformative negation of a statistical hypothesis cannot be taken as corroboration, let alone proof of that hypothesis. A vast number of alternatives to the Null are compatible with the data if $p < 0.05$, and the test does not strongly predicate anything about the statistical hypothesis of interest and next to nothing about the complex network of hypotheses it came from. Second, the p-value is only a good approximation of the actual probability if the probability distribution underlying the observed phenomenon is both known and appropriate for the test, which is almost never the case, especially in corpus linguistics. In more complex models (see below), this extends to a requirement that the model be specified correctly with all sources of variance accounted for at the right place (informally speaking) within the model, and in the correct form. Third, a p-value threshold of 0.05 tolerates, in the long run and under the Null, one false alarm in twenty. In many cases such a low bar seems rather cavalier, especially as sample sizes rise and the test becomes ever more sensitive to minute effects (see below). These issues and many more arise from the fact that NHST is an unstable hybrid of two incompatible inferential philosophies, Fisher’s significance testing and the Neyman–Pearson theory of decisions between hypotheses, welded into a procedure neither founder would endorse (see Lehmann 2011 for a historical deep dive).

Deborah Mayo’s *severity* requirement makes the deeper problem underlying statistical confusion such as in NHST precise (Mayo 1996; Mayo & Spanos 2006; Mayo 2018). In its weak form: *data do not provide good evidence for a hypothesis if they come from a procedure with very little capacity to have detected the hypothesis’s falsity, had it been false* (paraphrased

¹A marginal but real debate about significance testing in corpus linguistics does exist (see, for example, Kilgariff 2005; Koplenig 2019).

from Mayo & Spanos 2006: 162). A hypothesis earns the label *corroborated* only by surviving many tests with a real chance of exposing its potential falsehood. Reading a single low p-value, derived from one Null, as corroboration of the whole network of substantive and particular hypotheses behind it (Section 1) is a flagrant breach of that requirement: such a test has almost no capacity to detect errors in the hypotheses that actually matter.

It is crucial to keep in mind that *any* real divergence, however slight, makes the Null technically false. Even with moderate sample sizes, NHST detects this by throwing a low p-value. Suppose the effect posited in Study 1 held for one ten-thousandth of speakers. The Null would then be technically false, NHST would flag it given a large enough sample (or simply an unlucky one containing a text or two from those few speakers), and the result could be paraded as corroboration of Trudgill’s theory. It would be nothing of the kind: no severe test was performed, and a discrepancy that small tells us nothing about English under the Danelaw. This is also the mechanism behind the loose fit of Section 1: a low-severity test is precisely how an informal, underdetermined theory manufactures the confirmation we argued it could never legitimately earn. The fault is not in frequentist inference but in this use of it.

Many of the deficiencies we document below stem from this adherence to NHST. We will argue in Section 4 that a few straightforward practices would have forestalled most of them. Among those practices are a careful, bias-resistant inspection of one’s own models, reviewers equipped to catch inferential and statistical errors, and relief from the pressure to report only positive findings. We do not suspect the authors of dredging or negligence. We are confident they acted in good faith, having seen the same confusions in many other researchers and in our own earlier work. But the field suffers if papers like Walkden et al. (2023) pass without methodological scrutiny.

2 Statistical models in corpus linguistics

The studies in Walkden et al. (2023) rest on statistical models of a particular kind. The errors we will document are not arcane, and any reader can learn to see them. This section builds the necessary intuitions from the ground up so that by the end the reader will be able to look at the central result of the case studies and judge for themselves whether they supports anything at all.

2.1 What a model is, and what it is for

In a corpus, we record single events of speakers/writers facing the same choice again and again. In medieval English (Section 3.1), the decision is whether to mark plural on a pre-nominal element or leave it bare. In Spanish (Section 3.2), the decision is between expressing a pronominal subject overtly or dropping it. Each instance is a yes-or-no. We want to know which factors *push* that choice one way or the other, and how strongly they do so. Relevant factors might be the animacy of a noun’s referent, the dialect region, the century. A statistical **model** is a disciplined guess at that relationship. It is a statement made by the researcher that the rate of “yes” depends on specific factors in some specified way.

The classic linguistic example is the English dative alternation (*give the book to her* versus *give her the book*). The outcome is binary (shifted or not). Candidate predictors include the animacy and length of the object (Bresnan & Ford 2010; Wolk et al. 2013). A model lets us ask how much animacy shifts the odds toward one variant, expressed as a number. That number, the **coefficient**, is usually what we are really after. As opposed to other fields, linguists are not actually invested in predicting the next sentence a speaker will produce, but rather in quantifying *what influences the choice, and by how much*. Estimating it is not a matter of belief or eyeballing. A piece of mathematics implemented in software, the **estimator**, finds the coefficient values that best fit the observed data (thorough introductions in Gelman & Hill 2006; Fahrmeir et al. 2013).

A terminological point that turns out to matter is that one does not “run a model” (Walkden et al. 2023: 558). The researcher *specifies* a model as a complex hypothesis about what controls the variation, and the estimator then *fits* it to the data. The distinction is not pedantry. It is the difference between a causal claim one is putting to the test and a number a program happened to print.

2.2 The original sin of destructive aggregation

Here is where the first case study (Section 3.1) goes wrong, before any model is fitted at all. The outcome is binary — each pre-nominal element either carries plural inflection or does not. But Walkden et al. (2023) do not model those individual choices. They first *aggregate* and collapse all the instances in a text (or region, or period) into a single derived number, a difference-score, and then model *that*.

Watch what this does to a tiny sample. Suppose one text contains five relevant pre-nominal elements, of which three are inflected, and a second text contains two, of which none are. Table 1 illustrates the toy dataset.

Table 1: A fictitious example data set illustrating the pitfalls of aggregation

Text	Element	Inflection
A	1	yes
A	2	yes
A	3	no
A	4	yes
A	5	no
B	1	no
B	2	no

Modelled directly, this is seven observations, each a genuine yes-or-no that a predictor can be tested against. Aggregated, Text A contributes the single number 0.6 and Text B contributes 0.0. We end up with two data points instead of seven outcomes. And those two data points lie about the data. Text B’s number, resting on two elements, carries exactly

the same weight in the subsequent model estimation as A's number, aggregating five actual data points. Even a text containing five hundred elementary data points would end up contributing a single number. The within-text variation, which is the very signal we are trying to model, has been erased. The real study does this at scale and digests thousands of individual outcomes into a few dozen numbers before modelling begins. The information destroyed in that step cannot be recovered by any sophistication downstream.

2.3 Modelling the outcome as it actually is

If we keep the data binary instead, the right tool follows naturally. We cannot use ordinary linear regression, whose predictions run off to plus and minus infinity, to model a quantity that must stay between 0 and 1 (a probability). A **Generalised Linear Model** (GLM) solves this by passing the linear combination of predictors through a *link function* that keeps predictions in range. For binary outcomes, a model with a so-called logit link and a binomial distribution is typically used (with better alternatives existing for special cases). In such models, the coefficient β for a predictor reads as how much that predictor shifts the log-odds of “yes” (see any introduction to statistics for linguists such as Baayen 2008; Gries 2021). The crucial gain over aggregation is that every single instance enters the model on its own terms, weighted by its own existence rather than dissolved into a group average.

One complication remains, and it is the one aggregation might be, in effect, an inadequate attempt to handle. The instances are not independent. Any number of elements from the same text (or the same author, the same region, and so on) usually shares common tendencies we have not modelled. Treating them as individual data points incorrectly pretends they are independent. A **Generalised Linear Mixed Model** (GLMM) addresses this by adding *random effects*. Random effects allow each text, speaker, and so forth to have their own baseline rates around the overall trend, so that clustered data are not mistaken for independent evidence (Gries 2015; Schäfer 2020). This is the standard, well-understood remedy. Aggregating the clusters away is not.

2.4 Evaluating models

Suppose we have two fitted models and we wish to compare them, find out which one is “better” in some meaningful way. Researchers’ reflex often is to reach for single-number summaries, which are not uninformative but need to be supplemented by further analyses to be interpreted competently. Take R^2 , advertised as the proportion of variance the model explains. A high R^2 does not guarantee a good model, and aggregation inflates it artificially: smoothing thousands of noisy binary outcomes into a few tidy averages removes exactly the variation a model is supposed to struggle with, so the model looks far more explanatory than it is. The number went up because the data were made easier, not because anything was understood. In general, we advise to interpret only stark contrasts in R^2 and only for models with the same underlying distribution function and link (see Mittlböck & Schemper 1996; Tjur 2009).

Prediction accuracy is the proportion of choices the model calls correctly. It is even more seductive and must always be read against the **base rate**, which is the accuracy reachable

by blindly always guessing the most frequent outcome. If 95% of pre-nominal elements are uninflected, a model that simply predicts “uninflected” every time scores 95% while having learned nothing whatsoever. Quoted without its base rate, an accuracy figure is not evidence, it is decoration. On the other hand, a model that reaches the same 95% by actually discriminating between the cases is a different thing entirely. However, the achievement is the discrimination, not the single number.

Precision and recall get at exactly that discrimination, and they split the question accuracy blurs into two. Precision is the share of correct hits among all the cases the model flags as occurrences (true positives over all predicted positives). Recall is the share of actual occurrences the model manages to catch (true positives over all real events). Crucially, the threshold that turns a probability (assigned by the model to each data point) into a yes-or-no call is not fixed at 0.5. We can slide it from 0 to 1 and track precision and recall across that range, often visualised as a pair of curves whose crossing point is a compact summary of how well the model really separates the classes. We are going to show such curves in our re-analysis of the published models later. A model that does nothing but predict the modal category lands its crossing near the base rate. A model that genuinely discriminates pushes it higher.

We have introduced the statistics tools we need for a re-analysis of the data from Walkden et al. (2023). The second case study (Section 3.2) reports, for the coefficient of its key predictor, $\beta = 0.63$ with $p = 0.049$ (Table 8), a result that cleared the conventional 0.05 threshold by one part in a thousand, computed on a model whose per-coefficient p-values are trustworthy only if the rest of the model is approximately right, and built on the aggregated dots we have just watched lie about the data. In the main part of this paper, we ask the relevant question: was this the output of a test with a real chance of failing? The next section answers it.

3 Data and inferences in Walkden et al. (2023)

3.1 Study 1: Number agreement in English

3.1.1 What was reported

Study 1 concerns the loss of plural agreement marking within the NP during the Middle English period, read as a reduction of redundancy when plural marking disappears on attributive adjectives and determiners. The hypothesis attributes this partly to the influence of Scandinavian languages between the eighth and eleventh centuries, and treats the differing pace at which determiners and adjectives shed their marking as itself possibly contact-related. The study examines determiners (called *quantifiers* throughout, though not all determiners are quantificational) and monosyllabic adjectives across “47 texts” (p. 551) totalling 4,366 data points (called “tokens” on p. 551).²

²We feel uncomfortable having to add this footnote, but without it our reanalysis would appear to be inconsistent. In the published paper, the authors claim 47 texts. A closer look at the published script where data are prepared and analysed shows that they had adjective data for 44 texts and determiner data for 46 texts. To see this, examine `length(levels(as.factor(ex.dataadj$Text)))` and `length(levels(as.factor(ex.dataquant$Text)))` at line 137 of their file `number.R` where data loading and

From a plot overlaying texts as dots on a map of England (their Figure 1), the authors argue that some texts show stronger loss of inflection on the determiner than on the adjective.³ Walkden et al. (2023: 552) claim that “this difference does not depend on the date of composition of the manuscript, but rather on the dialectal region. The majority of texts which show a difference between quantifiers and adjectives come from areas where Scandinavian contact was intense.” They estimate a linear model “with the difference in agreement between adjectives and quantifiers as our dependent variable and region as independent variable.” With the Southern dialects as the reference category (least Scandinavian contact), they report a significant effect for the North and East Midlands but not the West (Walkden et al. 2023: 552, where Footnote 8 was omitted):

The results showed a significant effect of region for the North ($\beta=0.39$, $SE=0.15$, $t=2.6$, $p=0.01$) and the East Midlands ($\beta=0.2$, $SE=0.10$, $t=2.09$, $p=0.04$), but not the West ($\beta=0.13$, $SE=0.13$, $t=1.0$, $p=0.3$). This geographical distribution suggests that contact with Scandinavian speakers played a relevant role in the loss of inflectional morphology, but equally importantly, it shows that concord in quantifiers was more affected than concord with adjectives.

The inferential logic is pure NHST: every result with $p < 0.05$ is called *significant*. We return below to why *significant* is not *meaningful*. First, though, there is a more fundamental problem with the dependent variable as the paper never states clearly how the “difference in agreement” was measured. It cannot be a direct observation, since it is called a *difference* computed per text. The data package resolves this. For each text the authors took (i) the proportion of determiners with plural agreement and (ii) the proportion of adjectives with plural agreement, and (iii) subtracted the second from the first. This per-text difference score is the model’s response variable. It is exactly the type of aggregation whose dangers we set out in Section 2, now realised on actual data.

Rather than rehearse the objection in the abstract, we plot the resulting data and let it expose itself. Figure 1 shows all 44 texts on the x-axis, sorted by difference score, with the score on the y-axis. We have added two pieces of information the paper does not reveal. The shape and colour of each dot encode its dialectal region (see legend) and the *size* of each dot is proportional to the total number of observations behind it (determiners plus adjectives), which ranges from 12 for *Edmund-m4* to 315 for *Purvey*. The vertical bar on each dot encodes a further property, namely how strongly determiners are overrepresented relative to adjectives in that text.⁴ The bar scale differs from the y-axis scale and long bars have been clipped to fit, so the printed numbers should be read for the full picture. *Edmund*

preparation is done. The source of this discrepancy appears to be that some texts contained no annotated adjectives whatsoever after the authors performed some cleanups not described in the published paper. However, they start with 48 (not 47) labels for the Text variable. We used the 44 texts referenced both in the adjective and the determiner data set in Figure 1 as we consider the difference score to be undefined in the two cases without adjectives. A difference of two ratios is undefined if one of the ratios is $0 \div 0$. In the models below, we use all 46 texts because random effects models handle levels with zero data points well.

³Including the borders of the Danelaw would have helped here. The visual pattern is in any case not obvious to us.

⁴A text naturally contains more determiners than adjectives; the point is that the ratio varies wildly between the chosen texts. The published paper hides these imbalances and consequently does not explain them.

contributes 2 adjectives and 44 determiners — 22 times more determiners than adjectives — whereas *Reynard* contributes 13 adjectives and 16 determiners, the reverse imbalance.

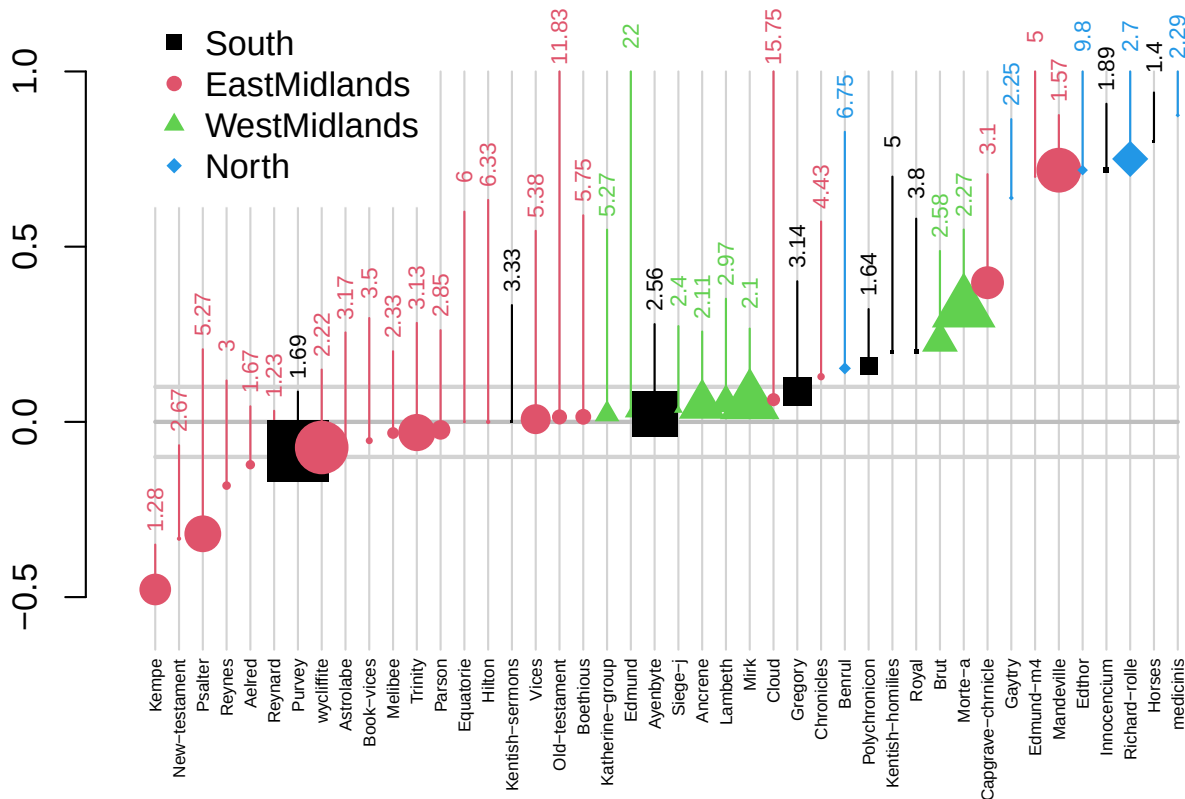


Figure 1: Difference scores for the texts in the sample; sizes of the dots correspond to the number of observations contributed by the text; vertical bars indicate the overrepresentation of determiners

The plot makes the damage concrete. The model behind the reported result treats the dot for the smallest text (12 observations) as exactly as informative as the dot for the largest (315 observations). A plain linear model on these scores is the wrong tool for aggregated proportions in the first place,⁵ and the imbalance compounds the problem. Were *Innocencium* and *Horses* as large as *Purvey* and *Ayenbyte*, the generalisations about the Southern group would weaken considerably. The difference score is itself an estimate resting on wildly varying sample sizes and adjective-to-determiner ratios, so it carries an inaccuracy the simple linear model is not built to absorb. Nominal significance levels almost certainly overstate the real ones. This is elementary enough that an author, reviewer, or editor should have caught it.⁶

⁵Models for proportion data exist, but they are harder to handle and interpret (Zuur et al. 2009). In any case the authors report a plain linear model, not a weighted binomial model on proportions.

⁶Whether more data have since been annotated, or the picture has since clarified, is beside the point we are making, although we honestly hope that it is the case. Walkden et al. (2023) as examined here is published, peer-reviewed research, and it had been downloaded 1,968 times as of 31 May 2026 according to the publisher's website.

Table 2: The Quasi-Reported Model is a binomial model with a logit link. $N=4366$, Tjur’s $R^2=0.147$. The prediction accuracy on the training data (at a probability threshold of 0.5) is 0.75 given a base rate of 0.71.

Term	Level	Coefficient	Sigma	P	Sig.
Intercept	(Adjective, South, 0)	1.96	10.72	<0.001	***
POS	Determiner	-0.38	1.78	0.075	
Dialect	WestMidlands	-0.72	3.40	0.001	***
	EastMidlands	-1.53	7.02	<0.001	***
	North	-0.53	1.55	0.122	
POS:Dialect	Determiner:WestMidlands	0.52	2.12	0.034	*
	Determiner:EastMidlands	0.08	0.33	0.739	
	Determiner:North	-2.52	6.47	<0.001	***

3.1.2 Alternative models

The unaggregated data are, fortunately, in the data package. The relevant linguistic event — even in the Middle Ages — was a single writer producing a single determiner or adjective, marked or unmarked. The right model therefore estimates the probability that plural agreement is overtly realised on a given word.⁷ We specified a model implementing the paper’s idea while avoiding the difference-score problem. In that model the response is the probability of overt agreement on a word, as a function of its part of speech (adjective or determiner), dialect group, and their interaction. We call it the **Quasi-Reported Model**, not what *was* reported, but what *should have been*. Table 2 gives the coefficients and p-values.

Granting a cavalier 0.05 threshold for the sake of argument, the estimates suggest that Midlands writers (East and West) drop agreement more readily overall, while Northern writers do not, but that Northern writers do show a relatively strong tendency to avoid agreement specifically on determiners (the POSDeterminer $\times$ DialectNorth interaction). That last point is in line with the paper. The overall dialect picture, however, is not: the West Midlands show an even stronger general tendency to avoid agreement than the East Midlands, which runs against the paper’s claim. We do not lean on this as a counter-finding as the West Midlands estimate rests on its own limited data. However, that is precisely the point. The dialect signal is too thin to support an inference in either direction. Read honestly, it is noise.

To be fair to the dialect predictors, dropping dialect from the specification yields a model that is worse by standard metrics such as R^2 , which indicates the regressors do carry some influence. We call this the Null Model (Table 3).

But the dots plot already warned us there are simply not many data from the North. That DialectNorth fails to reach significance on its own may reflect nothing more than scarce Northern data combined with a genuinely small effect. The plot also shows texts scattered

⁷The authors themselves follow this strategy in Study 2 (our Section 3.2), though there they introduce a different, equally inadmissible aggregated score as a regressor.

Table 3: The Null Model is a binomial model with a logit link. $N=4366$, Tjur’s $R^2=0.005$. The prediction accuracy on the training data (at a probability threshold of 0.5) is 0.71 given a base rate of 0.71.

Term	Level	Coefficient	Sigma	P	Sig.
Intercept	(Adjective)	1.15	16.86	<0.001	***
POS	Determiner	-0.35	4.53	<0.001	***

Table 4: Model comparison for the Null Model and the Quasi-Reported Model

Model	Tjur’s R^2
Null Model	0.005
Quasi-Reported Model	0.147

across the difference-score range regardless of dialect, with most clustered near zero; at the negative extreme, just five East Midlands texts account for every negative value. The honest response is to model variation at the document level as a random effect, absorbing differences in the circumstances, register, and individual style of each text. Table 5 reports the resulting **Text Model**.

The result is decisive: variation *between texts* swamps every other factor, and the model’s explanatory power rises sharply. The conditional $R^2=0.578$, against Tjur’s $R^2=0.147$ for the Quasi-Reported Model.⁸ The only generalisation surviving at the nominal threshold is the POSDeterminer $\times$ DialectNorth interaction — Northern writers avoiding agreement on determiners more than on adjectives. And that rests on 350 data points from 5 texts spanning 340 years. Where one must work with so little, it is worth asking whether statistics is the right instrument at all.

3.1.3 Inspection of the models

Beyond testing, a fitted binomial model also predicts, and the minimal requirement is that it predict its own input data with high accuracy. If it does, it differentiates at least somewhat between cases with and without agreement as seen in the data (external validity aside). For each combination of regressors it estimates a probability, and a threshold (conventionally 0.5) turns that into a binary call. We check the predictions on the very 4366 observations the model was fitted on. The Quasi-Reported Model (Table 6) predicts correctly in 3259 cases and incorrectly in 1107, or 74.6% — barely above the base rate of 70.8% (the accuracy of always guessing the modal outcome). It lands in that territory by overpredicting Agreement as the outcome. The Null Model (not tabulated) is weaker still. It always predicts overt agreement and so does not differentiate at all. In that regard, the Quasi-Reported Model is slightly better. However, the Text Model (Table 7) does markedly better, correct in 3740

⁸Comparison of R^2 measures is fraught with peril, as we pointed out in Section 2. However, a gap this large likely is no artefact.

Table 5: The Text Model is a multilevel binomial model with a logit link. $N=4366$, conditional $R^2=0.578$, marginal $R^2=0.155$. The prediction accuracy on the training data (at a probability threshold of 0.5) is 0.86 given a base rate of 0.71. The only random effect is Text with 46 levels ($s=3.3$).

Term	Level	Coefficient	Sigma	P	Sig.
Intercept	(Adjective, South, 0)	1.38	2.08	0.037	*
POS	Determiner	-0.50	1.99	0.047	*
Dialect	WestMidlands	1.13	1.46	0.145	
	EastMidlands	0.29	0.31	0.757	
	North	1.09	0.97	0.331	
POS:Dialect	Determiner:WestMidlands	0.16	0.55	0.580	
	Determiner:EastMidlands	-0.63	1.95	0.052	
	Determiner:North	-3.80	7.36	<0.001	***

Table 6: Correct and incorrect predictions of the Quasi-Reported Model

Predicted	Actual	
	Agreement	No Agreement
Agreement	3038	1056
No Agreement	51	221

cases and wrong in 626, or 85.7% prediction accuracy.

The same contrast appears if we vary the decision threshold from 0 to 1 rather than fixing it at 0.5. Figure 2 tracks accuracy across thresholds for both models. The Quasi-Reported Model sits flat at the base rate until a threshold near 0.8, then collapses to its complement. This is the signature of a model that is not really discriminating. The Text Model instead peaks near 0.5, as a well-behaved binomial model should, rising above the base rate because it genuinely separates the cases. Finally, precision and recall tell the same story from a third angle, see Figure 3.

Table 7: Correct and incorrect predictions of the Text Model

Predicted	Actual	
	Agreement	No Agreement
Agreement	2856	393
No Agreement	233	884

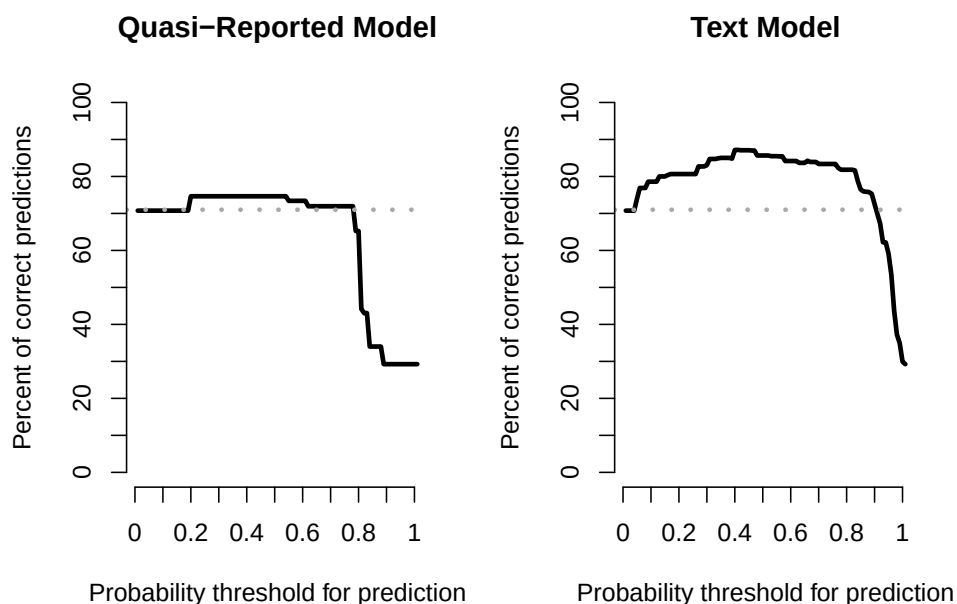


Figure 2: Percent of binary predictions based on the Quasi-Reported Model and the Text Model at varying thresholds; only for data used to estimate the model parameters and grouped by their actual outcome

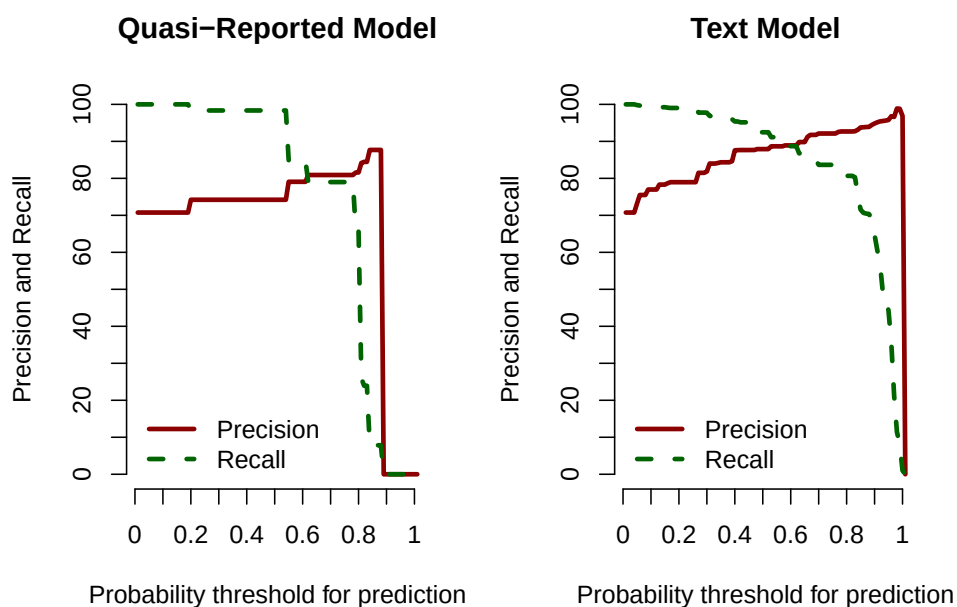


Figure 3: Precision and Recall for the Quasi-Reported Model and the Text Model at varying thresholds; only for data used to estimate the model parameters and grouped by their actual outcome

3.1.4 Final remarks

The reanalysis shows that the inferences in the paper are unwarranted. Once the inadmissible difference-score aggregation is removed, the data of Study 1 do not support the network of hypotheses of Section 1. The regressors' influence on the choice between marked and unmarked pre-nominal elements is so slight that only the standard NHST misreading of p-values could register it as relevant, and the dialectal signal is not merely small but internally contradictory. There is no clean North-East to South-West cline. Once the individual text enters as a random effect, the one factor with even moderate explanatory power is the text itself (also absorbing style and register) not dialect. Most importantly, including such a powerful random effect is not optional. Omitting it violates the independence-of-observations assumption the model rests on. Study 1 thus corroborates at best the philological insight that individual texts differ and deserve to be read individually.

A closing word on the line between exploratory description and inferential statistics is in order. A look at the data might hint — in some researcher's eye — that the North lost pre-nominal agreement more sharply than other regions, and that determiners there retained even less agreement than adjectives. To some, the data might look promising. We insist on proper model specification and evaluation anyway, not to spoil the result but because the whole purpose of statistics is to mark the boundary between (i) describing a sample, (ii) inferring to a population, and (iii) inferring to a causal network of hypotheses. Not every observed difference licenses the further steps. Whether it does depends on how much data there is and how large the difference.

Here the differences do not survive correct modelling at minimum for the reason that there are too few and too biased data. Of the $n = 4366$ data points, only 350 (about 8.02%) come from the North, while the West Midlands, where the per-text scores are most scattered, contribute by far the most (2063, or 47.25%). Worse, removing a single dominant Northern text, *Richard-rolle*, leaves just 165 Northern data points from four texts, or 3.95% of the reduced sample ($n = 4181$). Every claim about *the North* across several centuries thus rests on one text plus 165 further pre-nominal elements. Researchers are free to find these findings encouraging, but they cannot bear the weight of the inference placed on them in the case at hand. Blunt-force NHST inference and the difference-score aggregation are exactly what disguises this, collapsing 4366 data points into 44 (or 46 if you consider division by zero admissible arithmetic) and granting every text equal evidential weight regardless of size.

3.2 Study 2: Null subjects in Spanish

3.2.1 What was reported

Study 2 models the influence of orality, time, and region (Spain versus the Latin American countries) on the use of overt subjects in Spanish, against the language's strong overall preference for subject drop. Orality enters as the ORScore, an aggregate relative frequency of certain grammatical features in each text, thus a derived construct, not a direct observation. It is, in other words, the aggregated score we were promised back in Study 1 (Section 3.1), now doing duty as a *regressor* rather than a response. The original sin of Section 2 is committed here, too, on the other side of the equation. Table 8 gives the reported fit, where

Table 8: The Reported Model is a multilevel binomial model with a logit link. $N=3773$, conditional $R^2=0.206$, marginal $R^2=0.131$. The prediction accuracy on the training data (at a probability threshold of 0.5) is 0.89 given a base rate of 0.89. The only random effect is Text with 37 levels ($s=0.31$).

Term	Level	Coefficient	Sigma	P	Sig.
Intercept	(0, 0, Spain)	-2.55	9.24	<0.001	***
ORScoreZ		1.03	3.82	<0.001	***
YearZ		0.05	0.33	0.743	
Macroregion	Non-Spain	0.63	1.96	0.049	*
ORScoreZ:YearZ		-0.44	2.12	0.034	*

Subject is 1 for an overt subject, *ORScoreZ* the z-scored orality score, *YearZ* the z-scored document date, and *Macroregion* either Spain or Non-Spain. The random effect *Text* has 37 levels for 37 documents. We call this the Reported Model.

That our reconstruction matches the paper can be checked against the authors’ own report (Walkden et al. 2023: 558) (Footnote 14 omitted). The numbers line up, so we have specified the same model they did (highlighting ours).⁹

However, this trend also holds for Spain, which contradicts our sociolinguistic predictions. Since the orality effect discussed in section 4.3 may be at play in the bar charts, it is necessary to run a mixed-effects model to confirm any relationships across countries and centuries. A mixed model using *glmer* from the *lme4* package (Bates et al. 2015) in R (R Core Team 2021) was run to find any such underlying patterns. The model had pronoun realisation (3,773 total tokens) as the binary-dependent variable; macroregion (Spain vs. Non-Spain), orality and year (both z-scored) as fixed effects; and document ID as a random effect. Country was originally included as a fixed effect but had too many levels and was replaced with macroregion. The interaction between year and orality was significant ($\beta=-0.44$, $SE=0.21$, $p<0.03$), where the negative coefficient means that for every year that passes, the effect of orality on overtness is lessened. Thus, there is an underlying diachronic effect, but as we can see from Figure 3, it is a small effect. Macro-region was also found to be just significant ($\beta=0.63$, $SE=0.32$, $p<0.05$), favouring overtness in every country but Spain. This result explains the apparent rise we saw for Spain in Figure 3 as just a product of orality: the later Spanish texts score higher in ORSCORE.

We turn to two smaller things first, then to the substantive one. The p-values here are reported as inequalities (<), whereas Study 1 reported them as equalities (=), see Section 3.1.1. In both studies they are given with identical precision (two places), so the switch is puzzling. Also, the claim that it is “necessary to run a mixed-effects model to confirm any relationships” overstates the case. There is no theoretical necessity to include anything as

⁹After double- and triple-checking, we take the $p < 0.03$ reported for ORScoreZ:YearZ to be a minor lapse.

a random effect instead of a fixed effect (see Schäfer 2020; Gelman & Hill 2006: 245–247). However, it is definitely a good idea to model any known source of variance in the model, be it as a fixed or a random effect. The inference throughout is the NHST reflex described in Section 1.5. If $p < 0.05$, infer the effect. If $p \geq 0.05$, do not. It is applied here to a Macroregion coefficient that clears the line by the barest margin (0.049) and is then read as *just significant*.

The substantive problem is in the authors’ own gloss. They explain (p. 558) that the macroregion result “explains the apparent rise we saw for Spain in Figure 3 as just a product of orality: the later Spanish texts score higher in ORSCORE.” That is an admission that the model is misspecified. If Macroregion and ORScore covary, and the authors know they do — they say so — then leaving that covariation unmodelled is precisely the error. The two predictors are competing to explain the same variance, and the coefficients cannot be read as the paper reads them. An interaction, or some comparable move within the mixed-model framework, was called for. It is also unexplained why Macroregion was used in place of Country (see Section 3.2.2). But the point is moot, because the Study 2 model turns out to be even less informative than the Study 1 model, as we now show.

3.2.2 Alternative models

We start by setting the Reported Model against a Text-Only Model (not tabulated). Its structure consists of a fixed intercept plus a random intercept for text, and nothing else. In other words, it allows the baseline rate to vary from text to text while admitting no explanatory variables at all. The Text-Only Model is a multilevel binomial model with a logit link. $N=3773$, conditional $R^2=0.184$, marginal $R^2=0$. The prediction accuracy on the training data (at a probability threshold of 0.5) is 0.89 given a base rate of 0.89. The only random effect is Text with 37 levels ($s=0.74$). The marginal R^2 of the Text-Only Model is necessarily 0, since it has no fixed effects, yet the conditional R^2 of the Reported Model is only minimally higher. Even on this simple comparison, orality, macroregion, and date account for very little of the variance, as the R^2 measures and the prediction accuracy both show.

A second alternative is the Country-Genre Model (Table 9). It drops the supposed main regressor (orality) and the secondary one (year), putting country and genre in their place. Contrary to the statement in the paper, the standard maximum-likelihood estimator converges with no special handling, and the Country regressor does not have too many levels.¹⁰ Table 10 shows the Country-Genre Model to be marginally worse than the Reported Model. Why the authors thought the country regressor could not be added for technical reasons remains unclear: the estimator has no difficulty with the coefficients — they simply fail to reach 0.05.

Next, a more complex specification. The Full-Country Model (Table 11) is the Reported Model with orality and year retained, but country in place of macroregion and genre added.

On R^2 , the Full-Country Model edges ahead of the Reported Model (Table 12). However, the

¹⁰Even if it did, a random effect nesting document ID could and should have been used. Problems with the number of levels and the data points per level are among the practical (not theoretical) reasons for choosing either a fixed or a random effect (Gelman & Hill 2006; Schäfer 2020).

Table 9: The Country-Genre Model is a multilevel binomial model with a logit link. N=3773, conditional $R^2=0.185$, marginal $R^2=0.047$. The prediction accuracy on the training data (at a probability threshold of 0.5) is 0.89 given a base rate of 0.89. The only random effect is Text with 37 levels ($s=0.56$).

Term	Level	Coefficient	Sigma	P	Sig.
Intercept	(Spain, Nonliterary)	-2.79	7.62	<0.001	***
Country	Bolivia	0.32	0.64	0.521	
	Colombia	-0.12	0.28	0.780	
	Dominican Republic	0.02	0.04	0.967	
	Panamá	0.28	0.59	0.553	
Genre	Literary	0.27	0.86	0.388	
	Supplemental	1.31	2.53	0.012	*

Table 10: Model comparison for the Null Model and the Reported Model

Model	Conditional R^2	Marginal R^2
Country-Genre Model	0.185	0.047
Reported Model	0.206	0.131

Table 11: The Full-Country Model is a multilevel binomial model with a logit link. N=3773, conditional $R^2=0.21$, marginal $R^2=0.141$. The prediction accuracy on the training data (at a probability threshold of 0.5) is 0.89 given a base rate of 0.89. The only random effect is Text with 37 levels ($s=0.29$).

Term	Level	Coefficient	Sigma	P	Sig.
Intercept	(0, 0, Spain, DOC)	-2.62	8.54	<0.001	***
ORScoreZ		1.12	3.75	<0.001	***
YearZ		-0.06	0.33	0.740	
Country	Bolivia	0.96	2.05	0.040	*
	Colombia	0.68	1.61	0.107	
	Dominican Republic	0.39	0.99	0.321	
	Panamá	0.51	1.30	0.193	
Genre	Literary	0.20	0.78	0.437	
	Supplemental	0.28	0.56	0.579	
ORScoreZ:YearZ		-0.58	2.45	0.014	*

Table 12: Model comparison for the Full-Country Model and the Reported Model

Model	Conditional R^2	Marginal R^2
Full-Country Model	0.210	0.141
Reported Model	0.206	0.131

Table 13: The Full-No-Genre Model is a multilevel binomial model with a logit link. $N=3773$, conditional $R^2=0.211$, marginal $R^2=0.139$. The prediction accuracy on the training data (at a probability threshold of 0.5) is 0.89 given a base rate of 0.89. The only random effect is Text with 37 levels ($s=0.3$).

Term	Level	Coefficient	Sigma	P	Sig.
Intercept	(0, 0, Spain)	-2.52	9.25	<0.001	***
ORScoreZ		1.15	3.89	<0.001	***
YearZ		-0.04	0.23	0.818	
Country	Bolivia	0.99	2.14	0.032	*
	Colombia	0.76	2.02	0.044	*
	Dominican Republic	0.40	1.02	0.308	
	Panamá	0.53	1.36	0.174	
ORScoreZ:YearZ		-0.56	2.38	0.017	*

small shifts in conditional R^2 are just artefacts of the shifting marginal R^2 . More tellingly, country has exactly one level (Bolivia) far enough from the intercept (Spain) to reach significance. The paper’s conclusion that overtness is “favoured in every country but Spain” is therefore unwarranted even on NHST’s own terms. Drop genre, and the picture shifts again as in the Full-No-Genre Model, Tables 13 and 14.

Model specification, evidently, is not an exact art. Dropping genre pushes Colombia over the significance line as well, so the NHST verdict is now Spain, the Dominican Republic, and Panamá against Bolivia and Colombia. The R^2 , meanwhile, barely moves across all these specifications: it makes essentially no difference which regressors we include, the results stay middling at almost exactly the same level. The next section shows why, taking the Reported Model as one specimen of the many equally meaningless models the data admit.

Table 14: Model comparison for the Full-No-Genre Model and the Reported Model

Model	Conditional R^2	Marginal R^2
Full-No-Genre Model	0.211	0.139
Reported Model	0.206	0.131

3.2.3 Inspection of the models

As in Study 1, we check the Reported Model’s probability estimates on the data it was fitted on. At a 0.5 threshold it predicts a null subject in 3773 cases and a pronominal subject in 0. Indeed, it never predicts an overt subject at all. This follows from the uninformative regressors and the high base rate of 88.55% (no overt subject). To see whether a lower threshold would help, Figure 4 sets the Reported Model’s predictions beside the Text-Only Model’s as violin plots. The predictions are identically distributed to an incredibly high degree.

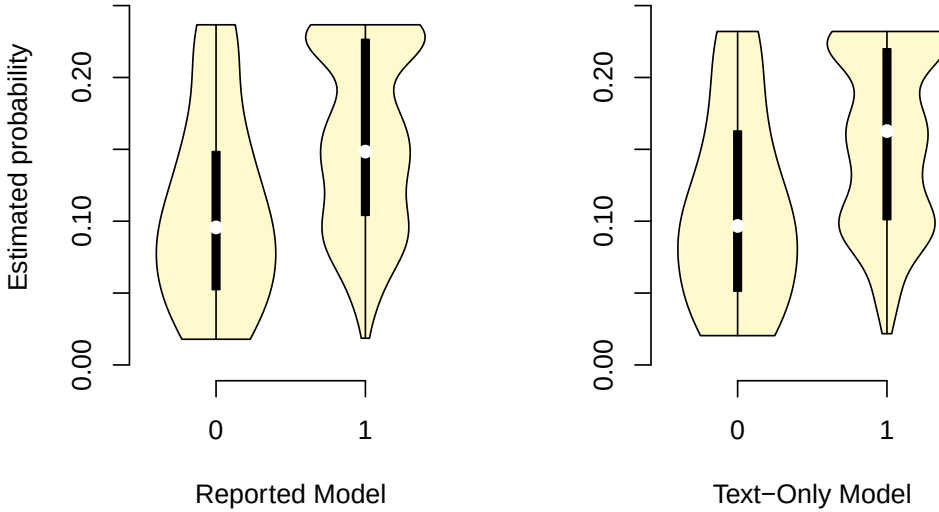


Figure 4: Probabilities estimated by the Reported Model and the Text-Only Model; only for data used to estimate the model parameters and grouped by their actual outcome

Even for the genuinely overt cases, the model’s estimated probability never exceeds 0.25. More striking still, the Reported Model’s distribution of predictions is all but identical to the Text-Only Model’s: orality, time, and country (or macroregion) are, by and large, irrelevant. Reaching significance is almost entirely a product of the simple model structure, minute regressor-correlated differences, and the fairly large sample. Those are exactly the conditions under which the critique of NHST from Section 1.5 applies most strongly. The left panel of Figure 5 plots correct binary predictions across thresholds; accuracy climbs quickly to 100% minus the base rate and stops, the signature of a model that does nothing but predict the modal category. The right panel gives precision and recall. Since the model never predicts a probability above 0.99, both fall to 0 beyond that threshold.

The clearest evidence comes last. Figure 6 plots the Reported Model’s predicted probabilities against the Text-Only Model’s, one dot per observed data point, correlation-plot style. Were the two models predicting identically, the dots would fall on a straight line — and that is essentially what we see, a straight line with negligible scatter. The two models carry virtually the same information: the mean absolute difference between their predictions is 0.0069, the maximum 0.045. We would like to point out that the 3773 data points collapse into 37 dots as the only effects in the model are the text random effect and — in the Reported Model — the additional fixed effects. Each document has — intrinsically given the model

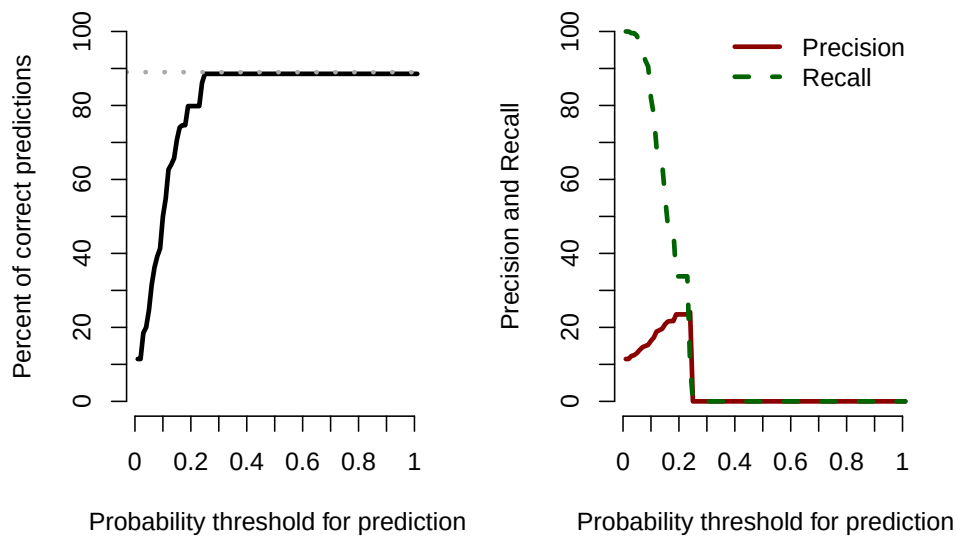


Figure 5: Left: Percent of binary predictions based on the Reported Model at varying thresholds; only for data used to estimate the model parameters and grouped by their actual outcome; Right: Precision and Recall for the Reported Model at varying thresholds; only for data used to estimate the model parameters and grouped by their actual outcome

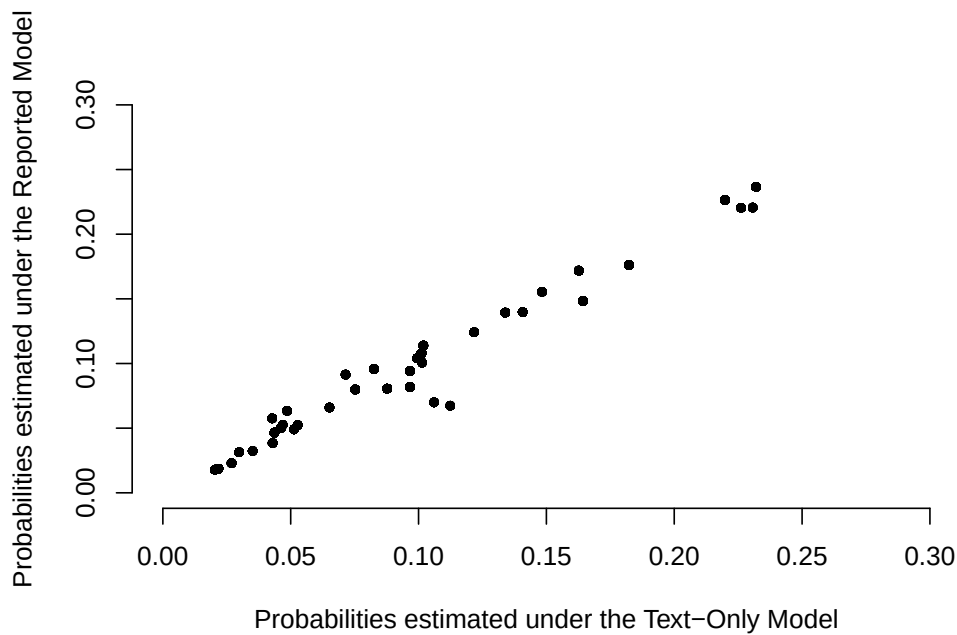


Figure 6: Comparing the predictions of the Reported Model and the Text-Only Model

structure — exactly one level/value per document. Hence, the model can only predict 37 different probabilities, no matter how many data points are contained in the texts. This is not per se a sign of an incorrectly specified model, it just defines a ceiling of informativity the model cannot exceed. Informative regressors would pull the 37 dots in Figure 6 away from the diagonal in some interpretable pattern, however.

In closing, we conclude that the data establish that individual Spanish texts differ from one another, and nothing more. They do not “confirm any relationships across countries and centuries.” With full access to the data these problems are not hard to find, which makes it the more regrettable that the analysis reached publication in this form. Negative findings are well worth publishing, and so are these data. But as negative findings resulting from an exploratory data analysis, not as corroboration of a large and loosely specified network of hypotheses, all the more so given the underdetermination described in Section 1.

We set Study 3 aside deliberately. It rests on generalised additive models, a distinct modelling apparatus that raises its own separate questions and would need a treatment of its own to do it justice, one this paper, focused on the GLM machinery of Studies 1 and 2, is not the place for. The closing section turns to what all of this means for the substantive hypotheses of Walkden et al. (2023) and for the field.

4 Putting data to good use

4.1 Failure mode

We hope to have shown that the data in Walkden et al. (2023) do not provide strong evidence for the authors’ claims, though the authors plainly believe they do. How did that happen? The shape of the mistake is by now familiar. A loosely specified theory (Section 1) was asked to draw support from a chain of derivations — substantive tenets to particular predictions to statistical hypotheses to Nulls of the form “*variable V has no influence*” — and a positive inference was then run all the way back up the chain, from “the Null is rejected” to “Trudgill’s theory is corroborated and our method works.” Each step treated every auxiliary hypothesis in the network as true and every rejected Null as a finding.

The chain breaks at both ends. At the top, the loose fit of an informal theory means even a clean rejection transmits almost nothing upward. The test had no real capacity to find the theory wrong, so passing it is not severity but coincidence (Section 1.5). At the bottom, the rejections themselves are unsound. They are products of NHST applied to aggregated, unweighted scores, so there is nothing to transmit in the first place. The result is not a modest finding, or a weak one, but no finding at all. We can only speculate why none of this was caught by the authors, the editors, or the reviewers. Rather than dwell there, we offer three remedies that would help, none of them ours, as other fields have met the same problems and thought hard about them.

4.2 Remedy 1: disciplined inference and better theories

Be modest about what an inference can carry. Testing is piecemeal, and any one study should aim at the most specific corner of the hypothesis network it can actually reach, not the grand theory above it. The deeper fix, though, is on the theory side. Linguistic theories would be far more testable if they predicted *numbers* or at minimum a numerical range the measurements should fall in. That is what sharpens a particular hypothesis into something a test can genuinely fail. Where a research tradition measures the same thing repeatedly, those ranges can also come from meta-analysis (see Vasishth 2015 for a rare example of meta-analysis in linguistics, albeit from psycholinguistics).

With numerical expectations in hand, one can sharpen tests using Neyman–Pearson logic (Neyman & Pearson 1933; Neyman 1937; Neyman 1957) or severity (Mayo & Spanos 2006) and earn a real positive inference, rather than merely rejecting an uninformative Null. But we should be honest about the catch. Principled confirmatory testing needs not only numerical predictions but data of sufficient quantity, control, and quality to put them at risk. Historical-corpus data of the kind Walkden et al. work with will, in all likelihood, never meet that bar. There simply are not, and never will be, enough well-controlled texts from the right places and centuries. Recognising when the data cannot bear *any* confirmatory framework is itself part of disciplined inference. The answer is not to run Neyman–Pearson or any approach to inferential statistics harder, but it is to stop forcing data into a method of inference not made for them.

And, maybe, the field also needs higher standards of statistical training. We should treat applied statistics as a core skill of the discipline, taught properly, rather than a bolt-on.

4.3 Remedy 2: open data as part of the review process

Reviewers and editors can only catch what they can see. A look at the underlying data should be a standard part of review, and reviewers should be equipped to make use of it. The clues in Walkden et al. (2023) as published are real but partial. The actual problems surface only on reanalysis, once one actually engages with the data set. In our experience also as authors, reviewers even at empirically minded journals rarely catch design or analysis errors, whether for want of the data or for want of the time to look. Behind this sits a missing standard. Most journals might still accept anything that reports a GLM (in the spirit of the diverse empirical and quantitative turns) without having established clear standards for the reporting of such statistical methods. Authors and reviewers can then impose whatever checks suit their taste. A journal with no best-practice guidelines for statistical methods arguably should not be publishing papers that lean on them at all, unless the paper’s own contribution is purely methodological and the paper is instrumental in establishing such guidelines.

4.4 Remedy 3: negative findings and preregistration

The pressure to publish, and to publish positive findings, is the often mourned underlying cause of the whole problem (Ioannidis 2005; Simmons, Nelson & Simonsohn 2011). One

answer to that is preregistration (Lakens 2019; Roettger 2021). The study’s motivation and design are submitted and judged before it is conducted, and the paper is published on the strength of the design whatever the results turn out to be. That makes negative findings publishable and gives researchers room to be hard on their own data. However, on its own preregistration might not be the perfect cure. Negative results draw few citations, so the incentive to find something publishable survives, as does the temptation to wave a good-looking result through without inspecting the data or the model. Recent work in other fields suggests preregistration has to be done exactly right to curb the practices the positive-results pressure encourages, see Brodeur et al. (2024).

We close on a constructive note rather than a sour one. Philology traditionally is a discipline of careful close reading, aimed at understanding texts in their linguistic and cultural setting. For good reason, it felt no obligation to the apparatus of the scientific method or to large-scale statistics. One reason was simply that the textual record is often too thin, across the relevant times and places, to support reliable statistical inference. Furthermore, individual sentences or even words in historic sources might require a careful interpretation (close reading) not recoverable with data-driven methods where sentences and words become mere data points. We ask readers to consider whether that has really changed. It is worth asking why a confirmatory inferential quantitative analysis should be reached for where a philological reading would serve better.

However, none of this is a brief against data analysis per se. Given the aforementioned problems with historical data sources, the distinction that matters is the one Study 1 already turned on. Describing a sample is one thing, inferring to a population another, and inferring to a causal network of hypotheses something else again. Exploratory data analysis is a genuine and valuable tool, but its goals are orthogonal to any testing framework, principled (like Neyman-Pearson) or otherwise (NHST). It asks what is in the data, not whether a theory survives. Read that way, the data of Walkden et al. (2023) should be put to good use. What is problematic about the paper is not the data but the confirmatory burden placed on them. We should match the instrument to what the data can bear, be it close reading, exploration, or, where the data genuinely allow, a severe test.

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References

- Baayen, R. Harald. 2008. *Analyzing linguistic data: A practical introduction to statistics using R*. Cambridge: Cambridge University Press. <https://doi.org/10.1017/cbo9780511801686>.
- Bresnan, Joan & Marilyn Ford. 2010. Predicting syntax: Processing dative constructions in American and Australian varieties of English. *Language* 86(1). 168–213. <https://doi.org/>

- 10.1353/lan.0.0189.
- Brodeur, Abel, Nikolai M. Cook, Jonathan S. Hartley & Anthony Heyes. 2024. Do preregistration and preanalysis plans reduce p-hacking and publication bias? Evidence from 15,992 test statistics and suggestions for improvement. *Journal of Political Economy Microeconomics* 2(3). 527–561. <https://doi.org/10.1086/730455>.
- Duhem, Pierre. 1914. *La théorie physique: Son objet et sa structure*. Paris: Marcel Rivière & Cie. <https://doi.org/10.4000/books.enseditions.6077>.
- Fahrmeir, Ludwig, Thomas Kneib, Stefan Lang & Brian Marx. 2013. *Regression – models, methods, and application*. Berlin: Springer. https://doi.org/10.1007/978-3-642-34333-9_2.
- Gelman, Andrew & Jennifer Hill. 2006. *Data analysis using regression and multilevel/hierarchical models*. Cambridge: Cambridge University Press. <https://doi.org/10.1017/cbo9780511790942>.
- Gigerenzer, Gerd. 2004. Mindless statistics. *The Journal of Socio-Economics* 33. 587–606. <https://doi.org/10.1016/j.socec.2004.09.033>.
- Greenland, Sander, Stephen J. Senn, Kenneth J. Rothman, John B. Carlin, Charles Poole, Steven N. Goodman & Douglas G. Altman. 2016. Statistical tests, P values, confidence intervals, and power: A guide to misinterpretations. *European Journal of Epidemiology* 31. 337–350. <https://doi.org/10.1007/s10654-016-0149-3>.
- Gries, Stefan Th. 2015. The most underused statistical method in corpus linguistics: Multi-level (and mixed-effects) models. *Corpora* 10(1). 95–126. <https://doi.org/10.3366/cor.2015.0068>.
- Gries, Stefan Th. 2021. *Statistics for linguistics with R: A practical introduction*. 3rd edn. Berlin, New York: de Gruyter Mouton. <https://doi.org/10.1515/9783110718256>.
- Ioannidis, John P. A. 2005. Why most published research findings are false. *PLoS Medicine* 2(8). e124. <https://doi.org/10.1080/09332480.2005.10722754>.
- Kilgariff, Adam. 2005. Language is never, ever, ever, random. *Corpus Linguistics and Linguistic Theory* 1(2). 263–276. <https://doi.org/10.1515/cllt.2005.1.2.263>.
- Koplenig, Alexander. 2019. Against statistical significance testing in corpus linguistics. *Corpus Linguistics and Linguistic Theory* 15(2). 321–346. <https://doi.org/10.1515/cllt-2016-0036>.
- Lakens, Daniël. 2019. The value of preregistration for psychological science: A conceptual analysis. *Japanese Psychological Review* 62(3). 221–230. https://doi.org/10.24602/sjpr.62.3_221.
- Laudan, Larry. 1990. Demystifying underdetermination. In C. Wade Savage (ed.), *Scientific theories*, 267–297. Minneapolis: University of Minnesota Press. <https://doi.org/10.2307/2026601>.
- Laudan, Larry & Jarrett Leplin. 1991. Empirical equivalence and underdetermination. *Journal of Philosophy* 88(9). 449–472.
- Le Verrier, Urbain. 1859. Lettre de M. Le Verrier à M. Faye sur la théorie de Mercure et sur le mouvement du périhélie de cette planète. *Comptes rendus hebdomadaires des séances de l'Académie des sciences* 49. 379–383.
- Lehmann, Erich L. 2011. *Fisher, Neyman, and the creation of classical statistics*. New York: Springer.
- Mayo, Deborah G. 1996. *Error and the growth of experimental knowledge*. Chicago: University of Chicago Press.

- sity of Chicago Press. <https://doi.org/10.7208/chicago/9780226511993.001.0001>.
- Mayo, Deborah G. (ed.). 2009. *Error and inference: Recent exchanges on experimental reasoning, reliability, and the objectivity and rationality of science*. Cambridge: Cambridge University Press.
- Mayo, Deborah G. 2018. *Statistical inference as severe testing: How to get beyond the statistics wars*. Cambridge: Cambridge University Press.
- Mayo, Deborah G. & Aris Spanos. 2006. Severe testing as a basic concept in a Neyman–Pearson philosophy of induction. *The British Journal for the Philosophy of Science* 57. 323–357. <https://doi.org/10.1093/bjps/axl003>.
- Mittlböck, Martina & Michael Schemper. 1996. Explained variation for logistic regression. *Statistics in Medicine* 15(19). 1987–1997.
- Neyman, Jerzy. 1937. Outline of a theory of statistical estimation based on the classical theory of probability. *Philosophical Transactions of the Royal Society A* 236(767). 333–379. <https://doi.org/10.1098/rsta.1937.0005>.
- Neyman, Jerzy. 1957. “Inductive behavior” as a basic concept of philosophy of science. *Review of the International Statistical Institute* 25(1). 7–22. <https://doi.org/10.2307/1401671>.
- Neyman, Jerzy & Egon S. Pearson. 1933. On the problem of the most efficient tests of statistical hypotheses. *Philosophical Transactions of the Royal Society A* 231. 694–706. https://doi.org/10.1007/978-1-4612-0919-5_6.
- Quine, Willard Van Orman. 1951. Two dogmas of empiricism. In *From a logical point of view*, 20–46. 2nd edn. Cambridge: Harvard University Press.
- Roettger, Timo B. 2021. Preregistration in experimental linguistics: Applications, challenges, and limitations. *Linguistics* 59(5). 1227–1249. <https://doi.org/10.1515/ling-2019-0048>.
- Schäfer, Roland. 2020. Generalized linear mixed models. In Stefan Gries & Magali Paquot (eds.), *A practical handbook of corpus linguistics*, 535–561. Berlin, Heidelberg: Springer. https://doi.org/10.1007/978-3-030-46216-1_22.
- Simmons, Joseph P., Leif D. Nelson & Uri Simonsohn. 2011. False-positive psychology: Undisclosed flexibility in data collection and analysis allows presenting anything as significant. *Psychological Science* 22(11). 1359–1366. <https://doi.org/10.1177/0956797611417632>.
- Tjur, Tue. 2009. Coefficients of determination in logistic regression models—a new proposal: The coefficient of discrimination. *The American Statistician* 63(4). 366–372. <https://doi.org/10.1198/tast.2009.08210>.
- Trudgill, Peter. 2011. *Sociolinguistic typology: Social determinants of linguistic complexity*. Oxford: Oxford University Press.
- Vasishth, Shravan. 2015. *A meta-analysis of relative clause processing in Mandarin Chinese using bias modelling*. School of Mathematics and Statistics of the University of Sheffield Master’s thesis. <http://www.ling.uni-potsdam.de/~vasishth/pdfs/VasishthMScStatistics.pdf>.
- Walkden, George, Gemma Hunter McCarley, Raquel Montero, Molly Rolf, Sarah Einhaus & Henri Kauhanen. 2023. Sociolinguistic typology meets historical corpus linguistics. *Transactions of the Philological Society* 121(3). 546–567. <https://doi.org/10.1111/1467-968X.12275>.

- Wolk, Christoph, Joan Bresnan, Anette Rosenbach & Benedikt Szmrecsanyi. 2013. Dative and genitive variability in Late Modern English: Exploring cross-constructional variation and change. *Diachronica* 30(3). 382–419. <https://doi.org/10.1075/dia.30.3.04wol>.
- Zuur, Alain F., Elena N. Ieno, Neil Walker, Anatoly A. Saveliev & Graham M. Smith. 2009. *Mixed effects models and extensions in ecology with R*. Berlin: Springer. <https://doi.org/10.1007/978-0-387-87458-6>.